

DALTON, GA, USA

WMO: 722154

Lat: 34.722N

Lon: 84.869W

Elev: 216

StdP: 98.75

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 53885

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-5.0	-13.0	1.3	-2.2	-11.4	1.5	-1.4	9.0	1.1	8.3	2.1	2.6	340	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.5	33.9	24.0	32.7	23.9	32.2	23.7	26.5	30.9	25.5	30.4	24.8	29.8	2.0	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.9	28.5	24.0	19.4	27.9	23.0	18.2	26.6	84.3	31.4	79.1	30.5	76.2	30.2	32.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.7	5.6	DB	-10.5	35.2	2.4	1.6	-12.2	36.4	-13.6	37.3	-14.9	38.3	-16.6	39.4	(4)
(5)			WB	-11.0	27.2	2.4	1.9	-12.7	28.6	-14.1	29.7	-15.5	30.8	-17.3	32.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	4.6	7.1	11.1	16.0	20.0	24.3	25.6	25.6	22.1	16.0	10.3	6.5	(6)	
	DBStd	8.45	5.50	5.02	4.78	4.17	3.59	2.25	1.97	2.01	3.06	4.50	4.57	4.77	(7)	
	HDD10.0	529	184	107	45	4	0	0	0	0	0	7	49	133	(8)	
	HDD18.3	1803	425	316	228	92	26	0	0	0	7	98	243	368	(9)	
	CDD10.0	2652	17	26	78	185	311	428	483	484	364	193	59	23	(10)	
	CDD18.3	882	0	1	3	22	78	178	225	226	121	26	2	1	(11)	
	CDH23.3	7875	0	4	37	251	711	1552	1994	2076	997	236	16	0	(12)	
	CDH26.7	3100	0	0	2	44	207	622	866	942	361	53	2	0	(13)	
(14) Wind	WSAvg	1.7	2.4	2.4	2.4	2.0	1.5	1.2	1.1	1.2	1.3	1.6	1.9	2.1	(14)	
(15) Precipitation	PrecAvg	1430	135	133	145	114	106	127	132	98	109	80	125	125	(15)	
	PrecMax	1811	260	317	269	244	336	188	240	160	216	194	253	210	(16)	
	PrecMin	859	66	49	63	47	15	34	39	24	25	6	68	51	(17)	
	PrecStd	245	48	64	59	49	66	46	57	33	52	55	49	46	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.9	23.7	27.0	30.0	32.5	35.0	35.1	36.2	33.9	31.1	26.0	21.3	(19)	
		MCWB	16.1	17.5	17.4	20.4	23.5	25.6	24.9	22.8	20.8	22.3	18.2	17.1	(20)	
	2%	DB	17.8	20.2	23.9	27.7	30.9	32.8	33.6	34.0	32.3	27.7	22.4	18.6	(21)	
		MCWB	14.6	15.3	16.3	19.1	22.4	24.4	24.9	23.7	21.6	20.0	16.6	16.4	(22)	
	5%	DB	16.2	17.7	22.2	26.3	28.9	32.1	32.5	32.7	30.9	26.1	20.9	16.6	(23)	
		MCWB	13.5	13.5	15.3	17.8	21.2	24.0	24.6	23.6	22.2	18.9	15.9	13.9	(24)	
	10%	DB	13.0	16.1	19.8	24.7	27.4	30.3	31.3	32.1	28.7	23.8	17.9	14.8	(25)	
		MCWB	10.7	11.7	13.9	17.5	20.3	23.1	24.2	23.5	21.6	18.0	14.2	12.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.3	18.5	19.0	22.7	25.7	28.4	29.2	27.1	25.0	23.3	21.1	19.3	(27)	
		MCDB	18.5	22.0	23.7	27.7	30.9	32.6	32.1	31.4	30.1	28.0	22.9	20.0	(28)	
	2%	WB	15.8	16.9	17.8	20.7	23.2	26.1	26.6	25.8	24.0	22.0	18.9	17.0	(29)	
		MCDB	17.4	19.2	21.3	25.5	28.0	30.5	30.6	31.1	28.5	25.7	21.1	17.9	(30)	
	5%	WB	14.0	15.0	16.6	19.5	22.2	25.0	25.5	25.1	23.2	20.8	17.2	14.8	(31)	
		MCDB	16.0	17.3	19.9	24.4	27.1	29.9	30.1	30.6	27.5	23.7	18.7	16.0	(32)	
	10%	WB	12.0	12.5	15.4	18.3	21.3	24.1	24.8	24.5	22.5	19.5	15.5	12.6	(33)	
		MCDB	13.4	14.7	18.9	22.6	25.7	28.6	29.6	29.5	26.3	22.7	17.6	14.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	10.9	11.8	12.9	12.0	11.3	10.9	11.5	11.7	12.8	12.2	9.8	(35)	
		MCDBR	12.0	14.2	15.1	15.2	14.2	13.4	12.6	13.4	14.4	15.0	14.8	11.7	(36)	
	5% WB	MCWBR	8.7	9.5	8.1	7.6	6.6	6.0	5.4	5.2	5.7	7.7	9.0	8.5	(37)	
		MCWBR	10.1	11.4	11.9	12.5	11.8	11.4	11.3	11.4	10.8	11.6	10.5	8.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.318	0.322	0.357	0.403	0.461	0.492	0.539	0.538	0.430	0.375	0.343	0.317	(40)	
		taud	2.503	2.487	2.417	2.280	2.152	2.102	1.989	1.976	2.263	2.413	2.446	2.528	(41)	
	Ebn at Noon		890	926	915	883	833	804	763	754	829	853	850	870	(42)	
		Edh at Noon	83	94	110	132	152	160	178	178	127	101	88	77	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.48	3.02	4.15	5.32	5.89	6.18	5.86	5.48	4.70	3.87	2.82	2.11	(44)	
	RadStd		0.18	0.39	0.36	0.34	0.45	0.45	0.41	0.30	0.44	0.48	0.32	0.22	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+145	+102

Nomenclature: See separate page